

have blown up a long time ago). It is an old, very old person with an impeccable memory. Mother Nature does not develop Alzheimer's—actually there is evidence that even humans would not easily lose brain function with age if they followed a regimen of stochastic exercise and stochastic fasting, took long walks, avoided sugar, bread, white rice, and stock market investments, and refrained from taking economics classes or reading such things as *The New York Times*.

Let me summarize my ideas about how Mother Nature deals with the Black Swan, both positive and negative—it knows much better than humans how to take advantage of positive Black Swans.

Redundancy as Insurance

First, *Mother Nature likes redundancies*, three different types of redundancies. The first, the simplest to understand, is defensive redundancy, the insurance type of redundancy that allows you to survive under adversity, thanks to the availability of spare parts. Look at the human body. We have two eyes, two lungs, two kidneys, even two brains (with the possible exception of corporate executives)—and each has more capacity than needed in ordinary circumstances. So redundancy *equals* insurance, and the apparent inefficiencies are associated with the costs of maintaining these spare parts and the energy needed to keep them around in spite of their idleness.

The exact opposite of redundancy is naïve optimization. I tell everyone to avoid attending (orthodox) economics classes and say that economics will fail us and blow us up (and, as we will see, we have proofs that it failed us; but, as I kept saying in the original text, we did not need them; all we needed was to look at the lack of scientific rigor—and of ethics). The reason is the following: It is largely based on notions of naïve optimization, mathematized (poorly) by Paul Samuelson—and this mathematics contributed massively to the construction of an error-prone society. An economist would find it *inefficient* to maintain two lungs and two kidneys: consider the costs involved in transporting these heavy items across the savannah. Such optimization would, eventually, kill you, after the first accident, the first “outlier.” Also, consider that if we gave Mother Nature to economists, it would dispense with individual kidneys: since we do not need them all the time, it would be more “efficient” if we sold ours and used a central kidney on a time-share basis. You could also lend your eyes at night since you do not need them to dream.

Almost every major idea in conventional economics (though a lesser